```
    "plannedGrossDuration": null,
    "plannedNetDuration": "32843",
    "transportMeans_code": null,
    "vehicle": null,
    "registrationCountry": null,
    "registrationNumber": null,
    "cargoVolume": 3.0,
    "volumeUoM": "M3",
    "cargoWeight": 300.0,
    "weightUoM": "KG",
    "cargoQuantity": 3.0,
    "quantityUoM": "EA",
    "departureLocationId": "VEN_DE",
    "departureLocationType_code": "LogisticLocation",
    "plannedDepartureDateTime": "2024-12-22T23:00:00+00:00",
    "plannedDepartureDateTimeZone": "CET",
    "arrivalLocationId": "SP_ZS01",
    "arrivalLocationType_code": "LogisticLocation",
    "plannedArrivalDateTime": "2024-12-22T23:00:00+00:00",
    "plannedArrivalDateTimeZone": "CET",
    "freightUnitTPs": [
      {
        "freightUnitAltKey": "xri://sap.com/id:LBN#10000000001:QX8CLNT912:FT1_FREIGHT_UNIT:4100070
        "lineNo": 1
      }
    ],
    "plannedTotalDistanceUoM": "KM",
    "stops": [
      {
        "locationId": "VEN_DE",
        "locationType_code": "LogisticLocation",
        "ordinalNo": 1,
        "stopId": "61000795220001"
      },
      {
        "locationId": "SP_ZS01",
        "locationType_code": "LogisticLocation",
        "ordinalNo": 2,
        "stopId": "61000795220002"
      }
    ],
    "shipmentType": "TOR",
    "shipperRefDocuments": [
      {
        "docId": "187024259",
        "docType_code": "58"
      },
      {
        "docId": "4500060293",
        "docType_code": "001"
      }
    ]
  }
```

Sending Events via Provider API for Order Events (GTT)

As a data contributor, you can send back the tracking data (that will later be converted to and processed as events in the GTT system) to a solution owner using the Provider API for Order Events (GTT). This chapter contains the following sections:

- Prerequisites and Configuration
- Monitoring Integration

Prerequisites and Configuration

Prerequisites

To perform the tasks described in this chapter, you, as a data contributor, must have the following prerequisites:

1. You have onboarded on SAP Business Network for Logistics.
2. You have accepted the invitation from the solution owner using the Manage Invitations app. For more, see the application help of Manage Invitations.

Configuration

You need to use the System Connections app to complete the API implementations. For details, see the Connectivity section of *Administration Guide for Logistics Providers*.

Related Information

Provider API for Order Events (GTT)

Monitoring Integration

On the SAP Fiori launchpad, pen the View Logs app. Open the Manage Integration Logs – Version 2.0 app from the **Quick Links** section to view all integration logs for solution owner and data contributor transactions.

For instructions on how to use the app, open the in-app help by selecting ⓘ within the **Manage Integration Logs – Version 2.0** app.

Generated Tenant-local Interfaces

This chapter describes how to use the generated tenant-local interfaces in SAP Business Network Global Track and Trace. These APIs are provided for integration between the GTT system and other systems, which allows solution owners to access GTT services and perform CRUD (create, read, update, delete) operations on the GTT system via API requests.

API Authentication

You can use the following to authenticate your API calls to GTT service interfaces:

- HTTP Basic Authentication: You must create a technical user and maintain it in the System Connections app. For more, see Maintain Technical User in the System Connections App.

Prerequisites

To use these APIs, you must fulfill the following prerequisites:

1. Configure API authentication method. For more, see API Authentication.
2. Get the following information required in the API request URI:
    - **Server Address**: You must provide the server address as the host in the request URI:
        - For **basic authentication**, server address is the URL of your GTT Fiori Launchpad. It varies from hosted data centers. Do the following to get the server address:
            a. Go to your subaccount that has subscribed Global Track and Trace
            b. Click **Services** > **Instances and Subscriptions** > **Global Track and Trace** > **Go to Application**
            c. After sign-in, you will see the URL.

            #### Example

            Here are example server addresses specific to data centers for basic authentication:

            - EU10: https:// *<subdomain>*.gtt.logistics.eu10.business-network.cloud.sap
            - US20: https:// *<subdomain>*.gtt.logistics.us20.business-network.cloud.sap
            - SA30: https:// *<subdomain>*.gtt.logistics.sa30.business-network.cloud.sap

            You must leave out “https://” when filing the information to an API request to avoid duplication.

    - **Model Information**: Prepare the following model information needed in the resource path of the request URI:
        - Model Namespace: It’s in the format of “com.<tenant name>.gtt.app.<model name>”. You can find the entire model namespace on the top of your target **Model Details** page.
        - Write Service Name: It’s in the format of “<model name>WriteService”.
        - Read Service Name: It’s in the format of “<model name>Service”.
        - Model Name: You can find it on the target model card.

        #### Example

        In this guide, the standard model "gttft1" is used as the sample model. The following model information is used as examples:

        - Model Namespace: com.lbngttapps.gtt.app.gttft1
        - Write Service Name: gttft1WriteService
        - Read Service Name: gttft1Service
        - Model Name: gttft1

After you finish the above tasks, you are ready to send API requests. In the sections followed, you can find examples of API request and response pair specific to different interface services.

IDOC Service

Description

- technical name: GTT_V2_EM_IDOC_Ingestion
- purpose: for ingesting EM IDOC. It's used for integration with SAP S/4HANA to send freight orders and freight bookings to SAP Business Network Global Track and Trace for real-time tracking.
- protocol: HTTP
- users: technical. For more, see the "Create a Technical User and Assign Role Collection" section in ***Administration Guide for Solution Owners.***
- same service used separately to:
    - Receive a Tracked Process EM IDOC
    - Receive an Event EM IDOC
    - Receive a Combined EM IDOC

For the IDOC predefined parameters, see IDOC PREDEFINED PARAMETERS.

Regarding how to generate IDOC, see How to Send Documents from SAP S/4HANA to SAP Business Network Global Track and Trace.

Receive a Tracked Process EM IDOC

Request

Request URI

- For HTTP basic authentication:

  https://<server address>/api/idoc/em/v1/TrackedProcess

Request Method

Post

Request Body

EM IDOC in XML format

Sample Code

```
<soap-env:Envelope xmlns:soap-env="http://schemas.xmlsoap.org/soap/envelope/">
    <soap-env:Header>
        <msgID:messageId xmlns:msgID="http://www.sap.com/webas/640/soap/features/messageId/">uu
    </soap-env:Header>
    <soap-env:Body>
        <EHPOST01 xmlns="urn:sap-com:document:sap:idoc:soap:messages">
            <IDOC BEGIN="1">
                <EDI_DC40 SEGMENT="1">
                    <TABNAM>EDI_DC40</TABNAM>
                    <MANDT>910</MANDT>
                    <DOCNUM>0000000002363293</DOCNUM>
```

```
                    <DOCREL>755</DOCREL>
                    <STATUS>30</STATUS>
                    <DIRECT>1</DIRECT>
                    <OUTMOD>2</OUTMOD>
                    <IDOCTYP>EHPOST01</IDOCTYP>
                    <MESTYP>AOPOST</MESTYP>
                    <SNDPOR>SAPQM7</SNDPOR>
                    <SNDPRT>LS</SNDPRT>
                    <SNDPRN>QM7CLNT910</SNDPRN>
                    <RCVPOR>GTTV2ACCTP</RCVPOR>
                    <RCVPRT>LS</RCVPRT>
                    <RCVPRN>GTTV2ACC</RCVPRN>
                    <CREDAT>20200423</CREDAT>
                    <CRETIM>115700</CRETIM>
                    <ARCKEY>urn:sap.com:msgid=0894EF4591011EDAA1A918CDC670A12C</ARCKEY>
                    <SERIAL>20200423115700</SERIAL>
                </EDI_DC40>
                <E1EHPAO SEGMENT="1">
                    <APPSYS>QM7CLNT910</APPSYS>
                    <APPOBJTYPE>YSHIPMENT_ACC</APPOBJTYPE>
                    <APPOBJID>0000008888</APPOBJID>
                    <E1EHPCP SEGMENT="1">
                        <PARAMNAME>FORWARDING_AGENT_NAME</PARAMNAME>
                        <PARAMINDEX>0000000000</PARAMINDEX>
                        <VALUE>VENDOR</VALUE>
                    </E1EHPCP>
        <E1EHPCP SEGMENT="1">
                       <PARAMNAME>ORIGINAL_SRC_SYSTEM</PARAMNAME>
                       <PARAMINDEX>0000000001</PARAMINDEX>
                       <VALUE>QM7CLNT999</VALUE>
                   </E1EHPCP>
                    <E1EHPCP SEGMENT="1">
                        <PARAMNAME>PARTITION_KEY_HINT</PARAMNAME>
                        <PARAMINDEX>0000000000</PARAMINDEX>
                        <VALUE>00001</VALUE>
                    </E1EHPCP>
                    <E1EHPCP SEGMENT="1">
                        <PARAMNAME>YN_SHP_HDR_DLV_NO</PARAMNAME>
                        <PARAMINDEX>0000000001</PARAMINDEX>
                        <VALUE>0080045176</VALUE>
                   </E1EHPCP>
                   <E1EHPCP SEGMENT="1">
                        <PARAMNAME>YN_SHP_HDR_DLV_NO_LOGSYS</PARAMNAME>
                        <PARAMINDEX>0000000001</PARAMINDEX>
                        <VALUE>QM7CLNT911</VALUE>
                   </E1EHPCP>
                   <E1EHPCP SEGMENT="1">
                        <PARAMNAME>PLANNED_TECHNICAL_TIMESTAMP</PARAMNAME>
                        <PARAMINDEX>0000000000</PARAMINDEX>
                        <VALUE>020201022000000</VALUE>
                   </E1EHPCP>
                   <E1EHPCP SEGMENT="1">
                        <PARAMNAME>PLANNED_TECHNICAL_TIMEZONE</PARAMNAME>
                        <PARAMINDEX>0000000000</PARAMINDEX>
```

```
                        <VALUE>CET</VALUE>
                    </E1EHPCP>
                     <E1EHPCP SEGMENT="1">
                        <PARAMNAME>REPORTED_BY</PARAMNAME>
                        <PARAMINDEX>0000000000</PARAMINDEX>
                        <VALUE>USER_NAME</VALUE>
                    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_TRXCOD</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>SHP_NO</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_TRXID</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>0000008888</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_APPSYS</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>QM7CLNT912</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_START_DATE</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>020200423115700</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_END_DATE</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>020200423115700</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_ACTION</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>A</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
              <PARAMNAME>E1EHPTID_TIMZON</PARAMNAME>
              <PARAMINDEX>0000000010</PARAMINDEX>
              <VALUE>CET</VALUE>
        </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_LOCID</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>LOCID001</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_LOCTYPE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>OneTimeLocation</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_TIMEZONE</PARAMNAME>
```

```
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>CET</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_LONGITUDE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>50.9</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_LATITUDE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>90.5</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_UNLOCODE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>unlocode</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_IATA_CODE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>iatacode</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_DESCRIPTION</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>test one time location</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_COUNTRY_CODE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>DE</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_CITY_NAME</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>City name 001</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_REGION_CODE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>Region code 001</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_HOUSE_NUMBER</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>1001</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_STREET_NAME</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>Test street</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
```

```
        <PARAMNAME>GTT_OTL_POSTAL_CODE</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>201201</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_EMAIL_ADDRESS</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>test@cc.com</VALUE>
    </E1EHPCP>
    <E1EHPCP SEGMENT="1">
        <PARAMNAME>GTT_OTL_PHONE_NUMBER</PARAMNAME>
        <PARAMINDEX>0000000001</PARAMINDEX>
        <VALUE>123456789</VALUE>
    </E1EHPCP>
                        <E1EHPEE SEGMENT="1">
                             <MILESTONENUM>0000000001</MILESTONENUM>
                             <MILESTONE>ARRIV_CARR</MILESTONE>
                             <CARRID>B01_FWAGNT</CARRID>
                             <LOCTYPE>SP</LOCTYPE>
                             <LOCID1>TM84</LOCID1>
                             <SNDID>B01_FWAGNT</SNDID>
                             <MSG_EXP_DATETIME>000000000000000</MSG_EXP_DATETIME>
                             <MSG_ER_EXP_DTIME>000000000000000</MSG_ER_EXP_DTIME>
                             <MSG_LT_EXP_DTIME>000000000000000</MSG_LT_EXP_DTIME>
                             <EVT_EXP_DATETIME>000000000000000</EVT_EXP_DATETIME>
                             <EVT_ER_EXP_DTIME>000000000000000</EVT_ER_EXP_DTIME>
                             <EVT_LT_EXP_DTIME>000000000000000</EVT_LT_EXP_DTIME>
                             <EVT_EXP_TZONE>CET</EVT_EXP_TZONE>
                             <ITEMIDENT>29JlHP417jgXgHZDnd01B0</ITEMIDENT>
                        </E1EHPEE>
                        <E1EHPTID SEGMENT="1">
                             <TRXCOD>SHP_NO</TRXCOD>
                             <TRXID>0000008888</TRXID>
                             <APPSYS>QM7CLNT912</APPSYS>
                             <START_DATE>020200423115700</START_DATE>
                             <END_DATE>020200423115700</END_DATE>
                             <ACTION>A</ACTION>
                             <TIMZON>CET</TIMZON>
                        </E1EHPTID>
                    </E1EHPAO>
                </IDOC>
            </EHPOST01>
        </soap-env:Body>
    </soap-env:Envelope>
```

ℹ Note

You can use association field name plus _LOGSYS to send a custom source system value for association field, i.e YN_SHP_HDR_DLV_NO_LOGSYS.

You can use APPSYS to send a custom source system value for reference.

Response

Response Codes

The following HTTP response codes may occur:

Code	HTTP Description	Track and Trace Semantic
200	OK	EM IDOC is received successfully
400	Bad Request	Something is not correct in the payload
401	Unauthorized	There is no authentication passed with the request
403	Forbidden	The user does not have the right privileges
404	Not found	Request URI is not correct
500	Internal Server Error	An unspecified error occurred during request processing

Response Body

Message in XML format

Sample Code

```
<SOAP-ENV:Envelope xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:Header/>
    <SOAP-ENV:Body>
        <EHPOST01Response xmlns="urn:sap-com:document:sap:idoc:soap:messages">
            <IdocAssign>
                <TransferId xmlns="urn:sap-com:document:sap:idoc:soap:messages">0000000002705455
                <DbId xmlns="urn:sap-com:document:sap:idoc:soap:messages">0000000002705455</DbId
            </IdocAssign>
        </EHPOST01Response>
    </SOAP-ENV:Body>
</SOAP-ENV:Envelope>
```

Receive an Event EM IDOC

Request

Request URI

- For HTTP basic authentication

  https://<server address>/api/idoc/em/v1/Event

Request Method

POST

Request Body

EM IDOC in XML format

Sample Code

```
<soap-env:Envelope xmlns:soap-env="http://schemas.xmlsoap.org/soap/envelope/">
    <soap-env:Header>
        <msgID:messageId xmlns:msgID="http://www.sap.com/webas/640/soap/features/messageId/">uui
    </soap-env:Header>
    <soap-env:Body>
        <EVMSTA02 xmlns="urn:sap-com:document:sap:idoc:soap:messages">
            <IDOC BEGIN="1">
                <EDI_DC40 SEGMENT="1">
                    <TABNAM>EDI_DC40</TABNAM>
                    <MANDT>910</MANDT>
                    <DOCNUM>0000000002368243</DOCNUM>
                    <DOCREL>755</DOCREL>
                    <STATUS>30</STATUS>
                    <DIRECT>1</DIRECT>
                    <OUTMOD>2</OUTMOD>
                    <IDOCTYP>EVMSTA02</IDOCTYP>
                    <MESTYP>EVMSTA</MESTYP>
                    <SNDPOR>SAPQM7</SNDPOR>
                    <SNDPRT>LS</SNDPRT>
                    <SNDPRN>QM7CLNT910</SNDPRN>
                    <RCVPOR>GTTV2ACCEV</RCVPOR>
                    <RCVPRT>LS</RCVPRT>
                    <RCVPRN>GTTV2ACC</RCVPRN>
                    <CREDAT>20200428</CREDAT>
                    <CRETIM>104329</CRETIM>
                    <ARCKEY>urn:sap.com:msgid=0894EF4591011EDAA2A58AA1AF6843F4</ARCKEY>
                    <SERIAL>20200428104329</SERIAL>
                </EDI_DC40>
                <E1EVMHDR02 SEGMENT="1">
                    <EVTID>ARRIV_CARR</EVTID>
                    <EVTDAT>20200428</EVTDAT>
                    <EVTTIM>104300</EVTTIM>
                    <EVTZON>CET</EVTZON>
                    <E1EVMTID SEGMENT="1">
                        <TRXCOD>SHP_NO</TRXCOD>
                        <TRXID>0000001452</TRXID>
                    </E1EVMTID>
                    <E1EVMLID SEGMENT="1">
                        <LOCCOD>SP</LOCCOD>
                        <LOCID1>TM84</LOCID1>
                    </E1EVMLID>
                    <E1EVMREF02>
                        <APPSYS>QM7CLNT910</APPSYS>
                        <APPOBJTYPE>YSHIPMENT_ACC</APPOBJTYPE>
                        <APPOBJID>0000008888</APPOBJID>
                        <STARTDATE>20200723</STARTDATE>
                        <STARTTIME>180000</STARTTIME>
                        <ENDDATE>20200724</ENDDATE>
                        <ENDTIME>180000</ENDTIME>
                    </E1EVMREF02>
                    <E1EVMPAR SEGMENT="1">
                        <PARAM_NAME>ACTUAL_TECHNICAL_DATETIME</PARAM_NAME>
                        <PARAM_VALUE>020201026000000</PARAM_VALUE>
                    </E1EVMPAR>
```